Chapter 1: Revising Natural Laws

The first explanation I will consider offers the most conservative response to miracle testimony. It preserves the assumption that testimony can be credible by questioning whether the law of nature that has been apparently violated is truly exceptionless. What we call laws of nature may not be absolute rules, but rather generalizations drawn from limited experience that hold until new evidence prompts a revision. On this view, a reported miracle does not necessarily reflect a supernatural disruption, but rather the limits of our prior understanding; the apparent violation may stem from an incomplete observational record. Testimony, then, does not need to be dismissed. It may instead serve as a signal that our current account of a natural law is insufficient and must be updated in light of new evidence.

Relevant Critiques of Hume

Of Miracles has been the subject of much scrutiny from critics throughout the centuries between its publication and today. Upon its release, it was met with backlash over its blasphemous subject matter and exposed Hume to accusations of atheism that would plague him for the duration of career. Hume's work on religious topics was a factor in his being barred from positions or professorships on account of his religious views, or lack thereof. Critics attacked the essay for its religious skepticism but also for its logical gaps or inconsistencies, with non-religious readers present among the crowd that took issue. Not only did Hume's argument invite religious outrage, it was also met with disapproval from other atheist scholars owing to the logical shortcomings of its arguments.

In 1762, one major criticism of the work was put forth by George Campbell, a Presbyterian minister who was the first of many to accuse Hume's logic of circularity.[1] The argument against Hume's point summarizes Hume's argument as the following:

1. Testimony in support of the laws of nature is without exception.
2. Thus, the laws of nature are true.
3. According to testimony, miracles defy the laws of nature.
4. Thus, this testimony cannot be valid.

Campbell, as well as many critics to follow, took issue with the premises of Hume's argument. In his initial claims Hume presupposes the inviolability of the laws of nature based on a testimonial history of complete uniformity. At the same time, testimony is presented as the source of doubt in accounts of miracles. Campbell argues that if testimony can be used as a means to establish the uniformity of nature, it should not be discredited as evidence for miracles in the right circumstances. That is, Hume's distrust in testimony when it advocates for the existence of a miracle is contradicted by his acceptance of testimony as evidence for the laws of

[1] Wertheimer, Molly. *Philosophy & Rhetoric*, vol. 18, no. 3, 1985, pp. 189–93. *JSTOR*, http://www.jstor.org/stable/40237445.

nature. As explained previously, Hume establishes the laws of nature based on the consistent experience in accordance with these patterns. If the laws of nature describe true, complete uniformity, this consistency must extend back to the beginning of time. Our evidence that a given law of nature held true hundreds of years ago does not come from our senses, rather, it is based on the testimony of others. Thus, if the vast majority of our evidence for the laws of nature comes in the form of testimony, it seems reasonable that testimony would be accepted as evidence against them as well. In weighing belief in the laws of nature against belief in a reported miracle, one is ultimately choosing between two conflicting sets of testimony. It is not immediately clear why the testimony supporting natural uniformity should be privileged over the testimony that challenges it. Campbell's critique exposes this tension by raising the question: why assume that the outlying report is false, rather than reconsider whether the supposed uniformity has in fact been broken?

Earman on Hume's Definitions

Another criticism of Hume's circularity is explained by contemporary philosopher John Earman in his book *Hume's Abject Failure*. Earman identifies paradoxical logic in the definitions of miracles set forth by Hume in the framing of his argument.[2] Hume's two definitions for miracles are as follows:

1. "A miracle is a violation of the laws of nature."
2. "A miracle may be accurately defined, a transgression of a law of nature by a particular volition of the Deity, or by the interposition of some invisible agent."[3]

Earman's critique targets a related logical issue, this time embedded directly in Hume's definition of miracles. If Hume assumes that the laws of nature are exceptionless, then defining miracles as violations of those laws renders miracles logically impossible from the outset. The structure of the problem can be generalized as follows: if all As are Bs, except in cases of supernatural intervention, then it is no longer true that all As are Bs. Hume's second definition of miracles, which allows for rare exceptions caused by divine or invisible agents, undermines his claim that the laws of nature admit no exceptions. His formulation suggests that natural laws are both inviolable and violable, depending on the circumstances. This internal contradiction complicates his position and calls into question the coherence of his argument.

In his critique, Earman accuses Hume of failing to provide an avenue by which miracles could possibly exist, since they are defined such that they are impossible.[4] Hume cannot simultaneously claim that the laws of nature are fully uniform and also that a hypothetical miracle would not abide by them; one must be false in order for miracles to be feasible. Thus, his

[2] Earman, John, *Hume's Abject Failure: The Argument Against Miracles* (New York, 2000; online edn, Oxford Academic, 1 Nov. 2003).
[3] Earman, 9.
[4] Earman, 13.

definitions are unjustifiably partial to his stance, not allowing for the possibility of miracles in the first place.

The previous issue of historical occurrences as a means of setting probabilities arises in this paradox. Hume has determined the inviolability of the laws of nature based on the uniformity of past experience. The earlier discussion surrounding probabilities highlights the importance of nonzero probabilities in order for Bayesian belief updating to function properly. Once again, miracles are ruled out from the start, as they are assigned a prior probability of zero and thus rendered impossible by definition. Hume's method of interpreting historical occurrences to estimate likelihood of future occurrences becomes an issue here. The absence of historical violations cannot, on its own, justify the assumption that natural laws are inviolable, particularly given Hume's own skepticism toward historical testimony. This internal tension undermines his claim that such laws hold unconditionally, and in doing so, opens the door for miracles to be assigned a nonzero probability.

According to Hume's argument, laws of nature are the basis by which we can identify miraculous events; an event is only miraculous if it violates the type of uniformity that we consider to be inviolable. Campbell and Earman's arguments challenge this. Campbell in particular suggests the following: if a law of nature is contradicted by testimony, perhaps it should no longer be considered a law of nature at all.[5] After all, it is no longer upheld by completely uniform testimonial experience. In that case, the definition for a miracle becomes less clear. Now that the uniformity has been violated, perhaps this pattern should no longer be considered a law of nature. Thus, the reportedly miraculous event does not actually violate any inviolable law. In this way, miracles become self-contradictory; if a miracle causes us to reject a previously accepted law of nature, it is no longer a miracle. It would more so be evidence against a frequently-occurring pattern. Hume's definition thus becomes highly problematic; any miracle would immediately deem itself non-miraculous. In this way, Hume's argument against miracles does not work, but it also does not exactly translate into an argument *for* the existence of miracles.

If the laws of nature are able to be revised or retracted upon hearing witness testimony contradicting them, we would no longer have to so fiercely reject testimony that does not align with them. Thus, if we accept the existence of events that violate uniformities previously considered laws of nature, we no longer partake in Hume's inconsistent treatment of testimony; testimony can work both in support and opposition of the laws of nature, and shall not be restricted to only being considered valid when it supports the established laws.

A well-known example from Hume's *Of Miracles* captures this idea vividly. He recounts the story of an Indian prince who had never experienced freezing temperatures.[6] When told that water could become solid and strong enough to walk on, the prince found it impossible to believe. Based on his uniform experience, this conclusion was entirely rational. Yet we now know that the phenomenon of ice is not miraculous, but simply unfamiliar to someone who had

[5] Wertheimer, 190.
[6] Hume, 120.

only encountered water in warm climates. His disbelief was not a failure of logic, but of exposure. Had the prince accepted the testimony, he would not have been abandoning a law of nature, but he would have been updating it. The testimony challenged not the consistency of nature, but the incomplete account of it known by the prince. This story illustrates how testimony that appears to describe a violation may instead signal the need to revise what we take to be the laws of nature.

Note, this chapter does not aim to prove that miracles occur, but rather to explore how our understanding of natural laws might shift in light of compelling testimony. If the relevant uniformity is overturned, then the event ceases to be a miracle in Hume's strict sense. Regardless, that shift still carries philosophical consequences.

Revising Hume's Definitions

While this paradoxical logic weakens the coherence of Hume's argument, it does not necessarily negate it entirely. His initial standard for belief in miracles still stands: "If the falsehood of his testimony would be more miraculous, than the event which he relates; then, and not till then, can he pretend to command my belief or opinion."[7] This principle can be reexamined if we adopt a slightly modified understanding of miracles that allows them to possess nonzero probability. To make conceptual room for this possibility, we can consider two ways of revising Hume's definition: one that strengthens the status of natural laws, and one that loosens it. Each option resolves the contradiction differently and carries distinct consequences for how we understand the relationship between laws of nature and miraculous events.

Revision #1: The laws of nature are insusceptible to transgressions. Thus, even miracles do not transgress the laws of nature.

This revision strengthens the inviolability of natural laws by proposing that miracles, if they occur, do not violate the true laws of nature, but rather disrupt some less fundamental regularity. On this view, the laws of nature are still understood as absolutely exceptionless, and any apparent violation must be reclassified as a disruption of some imperfect regularity, perhaps one that had merely appeared lawlike. Miracles would therefore not transgress genuine natural laws. However, this revision faces two major problems. First, it directly contradicts a central element of Hume's argument: that miracles are, by definition, violations of the laws of nature. If miracles are instead defined as exceptions to weaker regularities, they are no longer miracles in Hume's sense, and the entire framework for evaluating their credibility must be reconstructed. This would move the discussion far from Hume's original formulation, making it unclear whether we are still engaging with his position at all.

Second, this revision does not resolve the deeper epistemological problem in Hume's account. Hume holds that the laws of nature are established through repeated observation and testimony affirming the same outcomes over time. But if our confidence in these laws depends

[7] Hume, 127.

on testimony, then we must treat testimony as a valid form of evidence. This rehashes the same tension within Hume's argument brought up by Campbell. Hume accepts testimony as sufficient to establish the uniformity of nature but rejects it when it supports the occurrence of a miracle. Unless we can provide some justification for the absolute inviolability of these laws beyond historical observation or testimony, this revision fails to avoid the circularity identified previously. As long as miracles are ruled out solely because they have never been observed, and as long as testimony is dismissed only when it contradicts accepted regularities, the argument remains epistemically problematic.

Revision #2: The laws of nature are violable, and miracles are events in which the laws of nature are violated.

This second possible revision of Hume's logic instead provides a looser definition of the laws of nature. Rather than upholding the invincibility of the laws of nature, it provides an avenue by which they can be broken, and this is with the occurrence of a miracle. This revision resolves the initial contradiction where miracles must violate something absolutely inviolable. It also relieves Hume's argument of a reliance on historical testimony to verify the inviolability of the laws of nature; in this modified version, we might expect the laws of nature to generally hold, but we are not completely shut off to the idea of them being violated, even if it would be for the first time in history. In light of these benefits, it seems that this revision successfully revolves the contradiction and allows Hume's argument to continue.

This revision, however, has the interesting consequence of removing the barrier between miraculous events and so-called *marvellous* events.[8] *Marvellous* is the descriptor Hume gives to events that are highly unlikely yet not miraculous. Although marvellous events occur rarely, Hume distinguishes them from miracles by noting that they remain within the bounds of expectation and are still presumed to occur from time to time. The looser definition of marvellous events deems them only improbable, in contrast to the impossible status Hume gives to miraculous events. This distinction places marvellous events outside the scope of Hume's strongest skepticism. Unlike miracles, which he believes cannot be supported by testimony due to their contradiction of uniform experience, marvellous events may be weakly supported by testimony because they fall within the realm of rare but possible occurrences. Within Hume's belief-updating framework, this distinction becomes clearer: for miracles, the prior probability is effectively zero, so no amount of testimonial credibility can raise belief. For marvellous events, the prior is low but nonzero, allowing testimony, even when limited, to increase the hearer's credence to some degree.

This analysis addresses the major tension in Hume's treatment of miracles: his definitions and probability assignments leave no room for belief updating, even in the face of credible evidence. By adopting a more flexible conception of natural laws, one that allows them to be revised in light of new and well-supported observations, we avoid the contradictions within

[8] Hume, 131.

Hume's framework while preserving a probabilistic standard for belief. According to this view, a reported miracle does not necessarily indicate that a law has been violated in the metaphysical sense. Instead, it may suggest that our current understanding of the law is mistaken or in need of refinement. However, this revision comes at a cost. It blurs the line between the miraculous and the merely improbable, collapsing the conceptual distance between miracles and marvellous events. If miracles are no longer defined by absolute impossibility, they begin to resemble extreme outliers rather than genuine violations of nature. Testimony, under these conditions, becomes relevant again; it does not serve as conclusive proof, but it may function as a form of evidence indicating that our account of the natural world and its laws requires updating. In loosening Hume's definitions, we preserve the logic of belief updating while creating conceptual space for events that challenge our current explanatory framework. The next chapter turns to a different interpretive possibility: that miracle reports reflect not revisions of natural law, but rather appeals to unknown or misunderstood natural causes.